

CS 215 — Assignment 4

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Question 1

$$y = g(x) = Bx + \nu, \quad x = h(y) = B^{-1}(y - \nu).$$

$$Dg(y) = \frac{\partial g(y)}{\partial y} = B, \quad |Dg(y)| = |B|$$

For any Borel set $y \subset \mathbb{R}^d$,

$$f_Y(y) = \frac{f_X(x)}{|Dg(y)|} = f_X(B^{-1}(y - \nu)) \frac{1}{|B|}.$$

The PDF of $X \sim \mathcal{N}(\mu, C)$ is

$$f_X(x) = \frac{1}{(2\pi)^{d/2} |C|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^\top C^{-1}(x - \mu)\right).$$

Plugging $x = B^{-1}(y - \nu)$ gives

$$f_Y(y) = \frac{f_X(x)}{|B|} = \frac{1}{(2\pi)^{d/2} |C|^{1/2} |B|} \exp\left(-\frac{1}{2}(B^{-1}(y - \nu) - \mu)^\top C^{-1}(B^{-1}(y - \nu) - \mu)\right).$$

Let $m = B\mu + \nu$. Then $B^{-1}(y - \nu) - \mu = B^{-1}(y - m)$, hence

$$(B^{-1}(y - m))^\top C^{-1}(B^{-1}(y - m)) = (y - m)^\top ((B^{-1})^\top C^{-1} B^{-1})(y - m).$$

Note that

$$(B^{-1})^\top C^{-1} B^{-1} = (BCB^\top)^{-1}, \quad |C|^{1/2} |B| = |BCB^\top|^{1/2}.$$

Therefore,

$$f_Y(y) = \frac{1}{(2\pi)^{d/2} |BCB^\top|^{1/2}} \exp\left(-\frac{1}{2}(y - m)^\top (BCB^\top)^{-1}(y - m)\right).$$

Hence $Y \sim \mathcal{N}(B\mu + \nu, BCB^\top)$.

Question 2

Let $\{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^d$ with sample mean

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i, \quad \text{and} \quad \mathbf{Z} = [\mathbf{x}_1 - \bar{\mathbf{x}} \quad \mathbf{x}_2 - \bar{\mathbf{x}} \quad \cdots \quad \mathbf{x}_n - \bar{\mathbf{x}}] \in \mathbb{R}^{d \times n}.$$

Write components as $(\mathbf{x}_k)_r$ for the r -th entry of $\mathbf{x}_k$, and $\bar{x}_r$ for the r -th entry of $\bar{\mathbf{x}}$. Then

$$Z_{rk} = (\mathbf{x}_k - \bar{\mathbf{x}})_r = (\mathbf{x}_k)_r - \bar{x}_r.$$

Consider the (r, s) -th entry of $\mathbf{Z}\mathbf{Z}^\top$:

$$(\mathbf{Z}\mathbf{Z}^\top)_{rs} = \sum_{k=1}^n Z_{rk} Z_{ks}^\top = \sum_{k=1}^n Z_{rk} Z_{sk} = \sum_{k=1}^n ((\mathbf{x}_k)_r - \bar{x}_r)((\mathbf{x}_k)_s - \bar{x}_s). \quad (1)$$

Now look at:

$$\hat{\mathbf{C}} = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$$

Its (r, s) -th entry is

$$\sum_{k=1}^n [(\mathbf{x}_k - \bar{\mathbf{x}})(\mathbf{x}_k - \bar{\mathbf{x}})^\top]_{rs} = \sum_{k=1}^n ((\mathbf{x}_k)_r - \bar{x}_r)((\mathbf{x}_k)_s - \bar{x}_s), \quad (2)$$

which matches (1) exactly. Therefore, every (r, s) -entry agrees, and hence

$$\boxed{\mathbf{Z}\mathbf{Z}^\top = \sum_{k=1}^n (\mathbf{x}_k - \bar{\mathbf{x}})(\mathbf{x}_k - \bar{\mathbf{x}})^\top = \hat{\mathbf{C}}}$$

Now consider $\mathbf{L} = \mathbf{Z}^\top \mathbf{Z} \in R^{n \times n}$. Let λ and $\mathbf{v}$ be an eigenvalue pair of $\mathbf{L}$ ($\lambda > 0$ since $\mathbf{L}$ is of form $\mathbf{A}\mathbf{A}^\top$ with $|\mathbf{A}| \neq 0$),

$$\mathbf{Z}^\top \mathbf{Z} \mathbf{v} = \lambda \mathbf{v}.$$

Define

$$\mathbf{u} = \frac{1}{\sqrt{\lambda}} \mathbf{Z} \mathbf{v} \in R^d.$$

Then

$$\hat{\mathbf{C}} \mathbf{u} = \mathbf{Z}\mathbf{Z}^\top \left(\frac{1}{\sqrt{\lambda}} \mathbf{Z} \mathbf{v} \right) = \frac{1}{\sqrt{\lambda}} \mathbf{Z} (\mathbf{Z}^\top \mathbf{Z}) \mathbf{v} = \frac{1}{\sqrt{\lambda}} \mathbf{Z} (\lambda \mathbf{v}) = \lambda \mathbf{u},$$

so $(\lambda, \mathbf{u})$ is an eigenpair of $\hat{\mathbf{C}}$.

Hence, the nonzero eigenvalues of $\hat{\mathbf{C}}$ and $\mathbf{L}$ coincide, and their eigenvectors are related by

$$\boxed{\mathbf{u} = \frac{\mathbf{Z} \mathbf{v}}{\sqrt{\lambda}}, \quad \mathbf{v} = \frac{\mathbf{Z}^\top \mathbf{u}}{\sqrt{\lambda}}}$$

Question 3

Directions of extremal variance: For a random vector X with mean μ covariance C and a fixed (unit) direction $v \in R^d$ we define the variance of X along the direction v as:

$$E[(v^\top (X - \mu))^2] = E[v^\top (X - \mu)(X - \mu)^\top v] = v^\top E[(X - \mu)(X - \mu)^\top] v = v^\top \mathbf{C} v.$$

Because C is symmetric non singular matrix, it has an orthonormal eigenbasis $\{u_i\}$ with $C u_i = \lambda_i u_i$.

For any unit vector v write $v = \sum_i \alpha_i u_i$ with $\sum_i \alpha_i^2 = 1$. Then

$$v^\top C v = \sum_i \lambda_i \alpha_i^2.$$

This is maximized (and minimized) when all of the weight is given to the eigen vector with maximum (and minimum) eigen value

$$\min v^\top C v = \lambda_{\min} \quad \text{and} \quad \max v^\top C v = \lambda_{\max},$$

achieved when $v = u_i$ and $v = u_j$, respectively (where i and j are such that λ_i and λ_j are minimum and maximum respectively)

(a) $\mu = (1, 1)^\top$, $C = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$

A valid factorization is $C = AA^\top$ with

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}.$$

Since C is diagonal, its orthonormal eigenvectors are the coordinate axes $u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, with eigenvalues $\lambda_1 = 9, \lambda_2 = 4$.

From the above derivation the variance along a unit direction v is maximized/minimized exactly when v is an eigenvector of C . Hence the extreme directions are u_1 (max) and u_2 (min), and the corresponding variances equal the eigenvalues:

$$\text{max-var dir: } v_{\max} = u_1, \text{ Var}_{\max} = \lambda_1 = 9, \quad \text{min-var dir: } v_{\min} = u_2, \text{ Var}_{\min} = \lambda_2 = 4.$$

(b) $\mu = (1, 2)^\top$, $C = \begin{bmatrix} 9 & 1 \\ 1 & 4 \end{bmatrix}$

A valid factorization is $C = AA^\top$ with $A = \begin{bmatrix} 3 & 0 \\ \frac{1}{3} & \frac{\sqrt{35}}{3} \end{bmatrix}$. and eigenvalues of C are

$$\lambda_{1,2} = \frac{13 \pm \sqrt{29}}{2},$$

Corresponding (unit) eigenvectors are

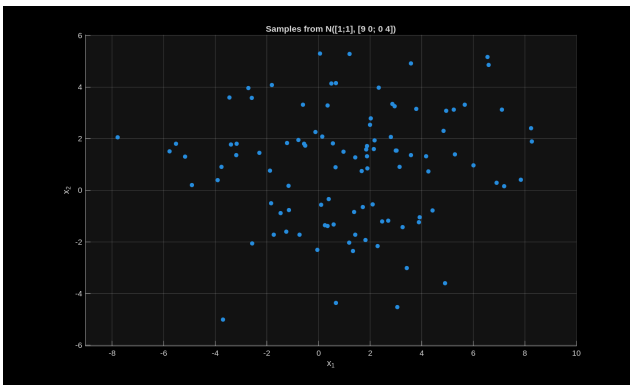
$$u_i = \frac{1}{\sqrt{1 + (\lambda_i - 9)^2}} \begin{bmatrix} 1 \\ \lambda_i - 9 \end{bmatrix}, \quad i = 1, 2,$$

Hence the extreme directions are u_1 (max) and u_2 (min), and the variances equal the eigenvalues:

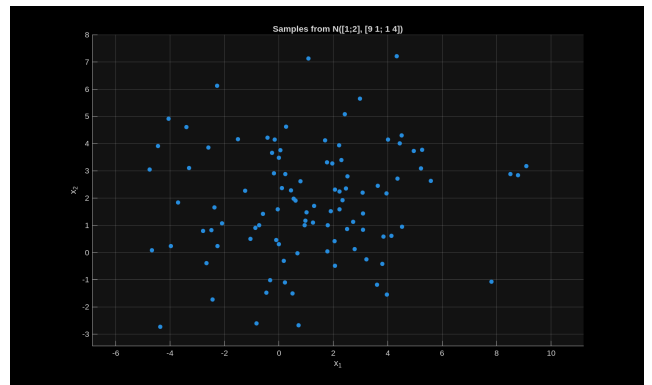
$$\text{max-var dir: } v_{\max} = u_1, \text{ Var}_{\max} = \lambda_1 = \frac{13 + \sqrt{29}}{2}, \quad \text{min-var dir: } v_{\min} = u_2, \text{ Var}_{\min} = \lambda_2 = \frac{13 - \sqrt{29}}{2}.$$

Experimental results: We now sample gaussian and find the max and min unit vectors experimentally.

Scatter plots (100 samples). Figure shows $n = 100$ samples from each Gaussian.



(a) Case (a): $\mu = (1, 1)^\top$, $C = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$.

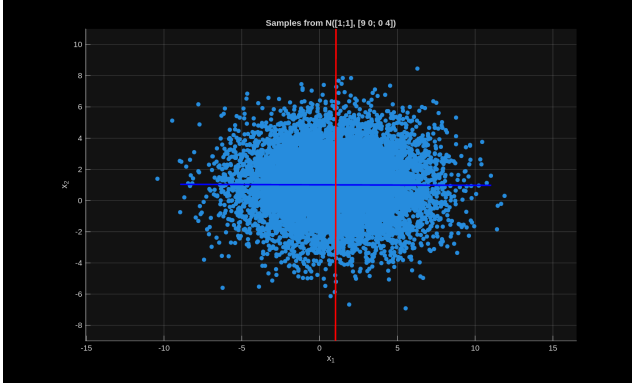


(b) Case (b): $\mu = (1, 2)^\top$, $C = \begin{bmatrix} 9 & 1 \\ 1 & 4 \end{bmatrix}$.

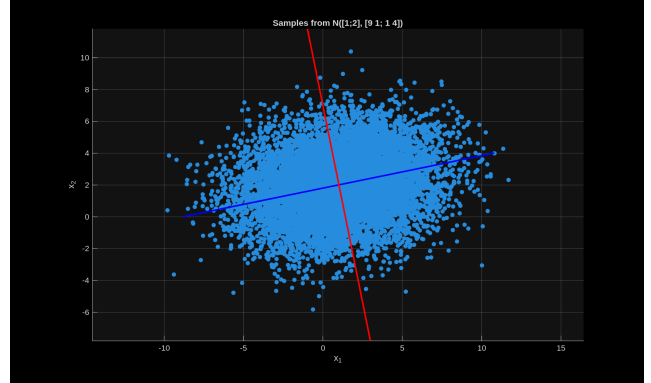
Figure 1: Scatter plots of $n = 100$ samples for the two covariance structures.

Extremizing directions: We find the directions using the above samples but using just 100 samples is not accurate enough, so we have sampled 10000 points and found the unit vectors which maximize and minimize the variance.

Overlays with 10,000 samples. Figure displays $n = 10,000$ samples for each case, with the max/min variance directions (experimental) overlaid as lines through the sample mean.



(a) Case (a) with 10^4 samples and extremal directions overlaid.



(b) Case (b) with 10^4 samples and extremal directions overlaid.

Figure 2: Empirically found extremizing directions.

Empirical extremizing directions:

$$\text{Case (a): } \hat{v}_{\max} = (-0.999995, 0.003145)^\top, \quad \hat{v}_{\min} = (0.004717, 0.999989)^\top$$

$$\text{Case (b): } \hat{v}_{\max} = (0.985616, 0.169001)^\top, \quad \hat{v}_{\min} = (-0.170550, 0.985349)^\top$$

Question 4

Let F_n and G_m be the empirical CDFs of two independent samples $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$. Define the test statistic

$$D_2 = \max_{x \in \mathbb{R}} |F_n(x) - G_m(x)|.$$

Fix $d \in [0, 1]$. Under the null hypothesis H_0 : both samples come from the same (continuous) CDF F ,

$$\begin{aligned} P_F(D_2 \leq d) &= P\left(\max_{x \in \mathbb{R}} |F_n(x) - G_m(x)| \leq d\right) \\ &= P\left(\max_{x \in \mathbb{R}} \left| \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\} - \frac{1}{m} \sum_{j=1}^m \mathbf{1}\{Y_j \leq x\} \right| \leq d\right) \\ &= P\left(\max_{x \in \mathbb{R}} \left| \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{F(X_i) \leq F(x)\} - \frac{1}{m} \sum_{j=1}^m \mathbf{1}\{F(Y_j) \leq F(x)\} \right| \leq d\right) \\ &= P\left(\max_{x \in \mathbb{R}} \left| \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{U_i \leq F(x)\} - \frac{1}{m} \sum_{j=1}^m \mathbf{1}\{U_j \leq F(x)\} \right| \leq d\right) \\ &= P\left(\max_{y \in [0,1]} \left| \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{U_i \leq y\} - \frac{1}{m} \sum_{j=1}^m \mathbf{1}\{U_j \leq y\} \right| \leq d\right) \end{aligned}$$

The right-hand side depends only on n, m , and the $\text{Unif}(0, 1)$ variables, and not on F . Therefore, under H_0 , the distribution of D_2 is independent of the underlying CDF F .

Question 5

Assumptions (When Null hypothesis is true)

- The paired differences in weekly loss, $d_i = \text{After}_i - \text{Before}_i$, are independently and identically distributed and follow a normal distribution
- All plants' differences come from the **same distribution**, i.e. they share a common variance σ_d^2 : $d_i \sim \mathcal{N}(0, \sigma_d^2)$.

We test whether the safety program reduced the average weekly loss in labor-hours using t -test.

Let $d_i = \text{After}_i - \text{Before}_i$ for $i = 1, \dots, 10$. From the table,

$$d = (-7.5, 2.5, -2.5, -3.5, -1.5, 0.5, 1, -4.5, -4.5, -1.5).$$

Hence

$$\bar{d} = -2.15, \quad s_d = 3.0005, \quad n = 10.$$

Hypotheses

$$H_0 : \mu_d = 0 \quad \text{vs} \quad H_1 : \mu_d < 0$$

Test statistic

$$t = \frac{\bar{d} - 0}{s_d/\sqrt{n}} = \frac{-2.15}{3.0005/\sqrt{10}} = -2.266, \quad \text{df} = n - 1 = 9.$$

p-value

$$p = \Pr(T_9 \leq -2.266) \approx 0.025$$

Decision at $\alpha = 0.05$ Since $p \approx 0.025 < 0.05$, reject H_0 . There is statistically significant evidence that the program reduced the average weekly loss in labor-hours; thus the safety program is effective at the 5% level.

Question 6

Part (c)

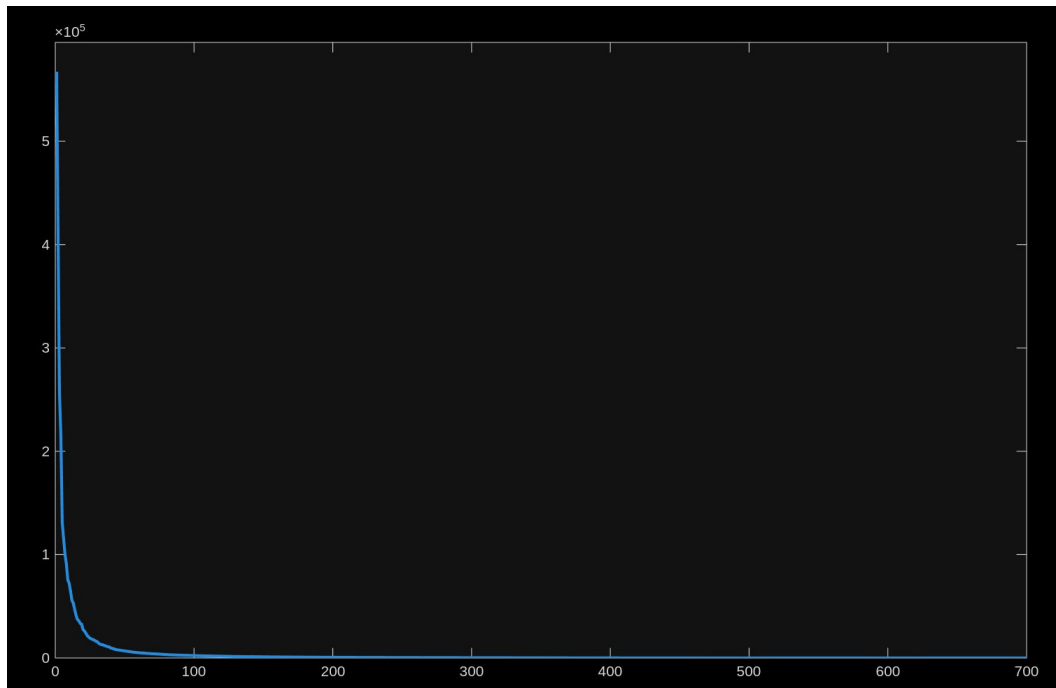
The eigenvalue graph for each digit initially drops sharply and then becomes nearly flat. The few number of high eigenvalues capture the high variances and the large number of low eigenvalues capture the low variances in the training data.

Let the location where the graph changes from the sharp drop to nearly flat be called the turning point. The position of the turning point depends on how complicated the number is. The simpler or more uniform the number is, the steeper the graph is, i.e it has an earlier turning point. For example, the turning point for 1 occurs earlier than that of 3, that is the graph of 1 is steeper than that of 3.

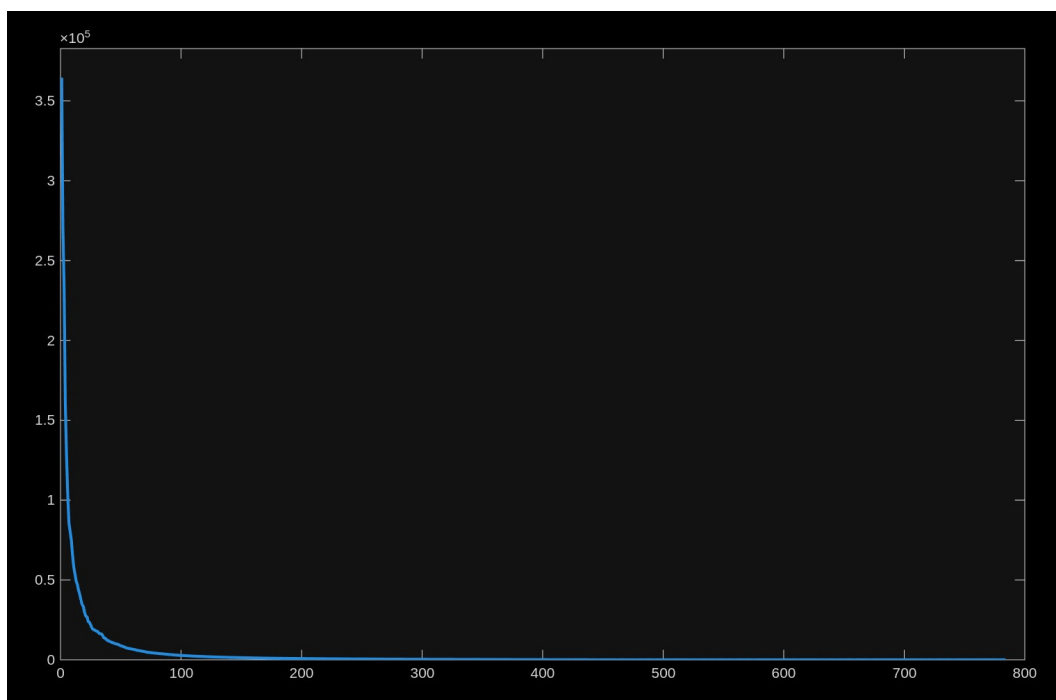
Part (d)

These images show where the intensity tends to increase or decrease relative to the class mean. The image of the first eigenvector shows the highest deformation showing the overall shape of the number. As we go from the second to the fifth eigenvectors, we can see the finer details like the smaller lines and curves of the number more clearly

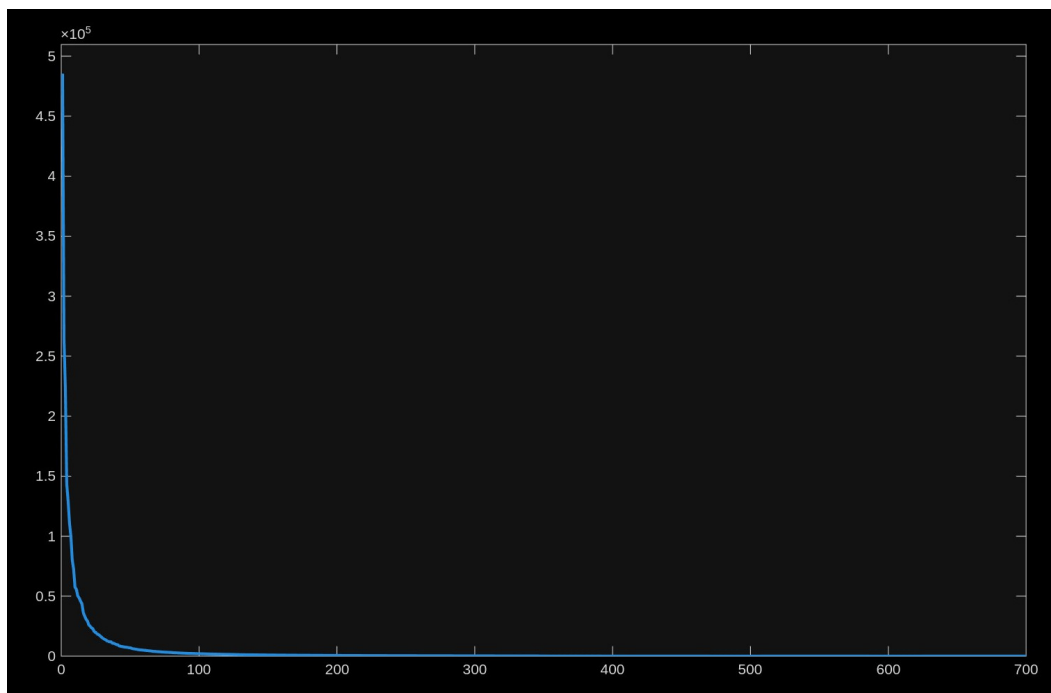
The eigenvalue plots for the digits 0,3,6 and 7:



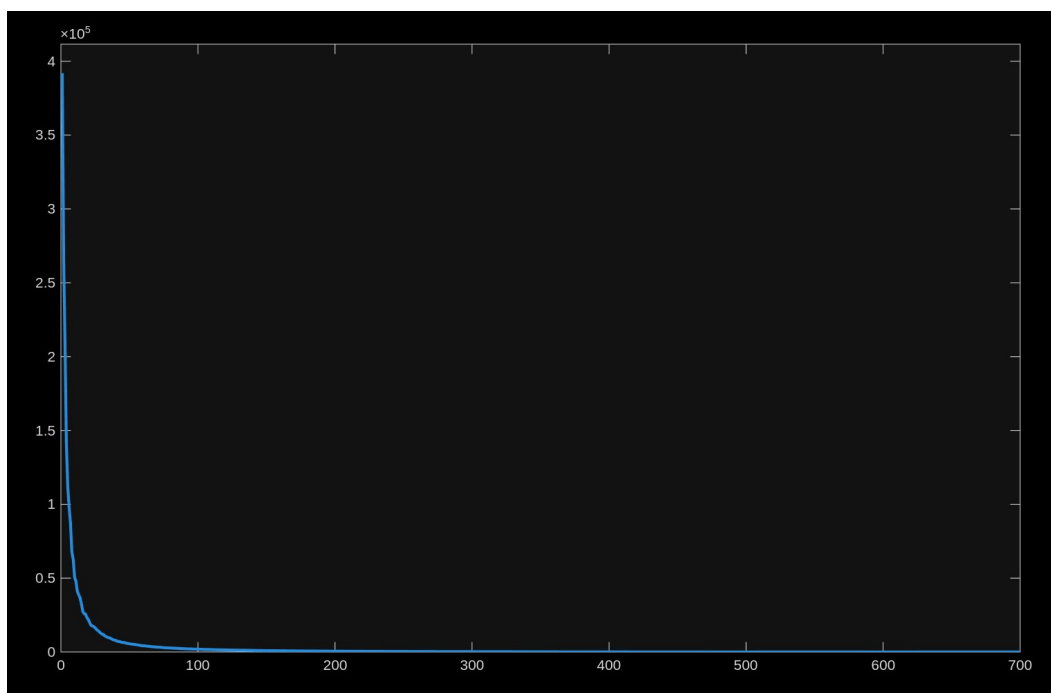
Digit 0



Digit 3

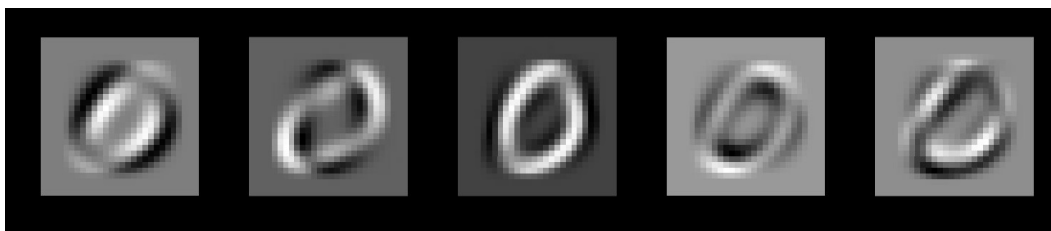


Digit 6

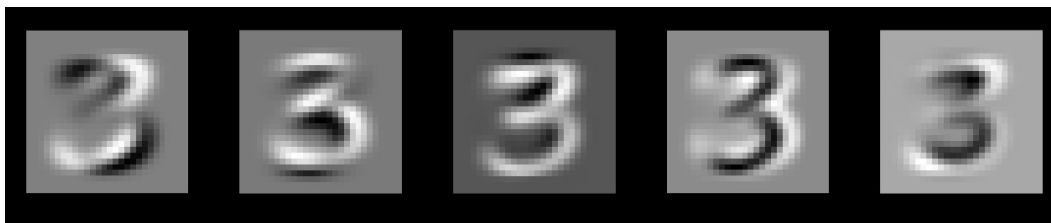


Digit 7

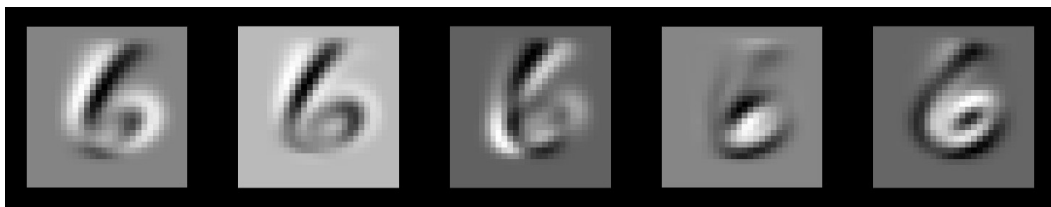
The eigenvector plots for the digits 0,3,6 and 7:



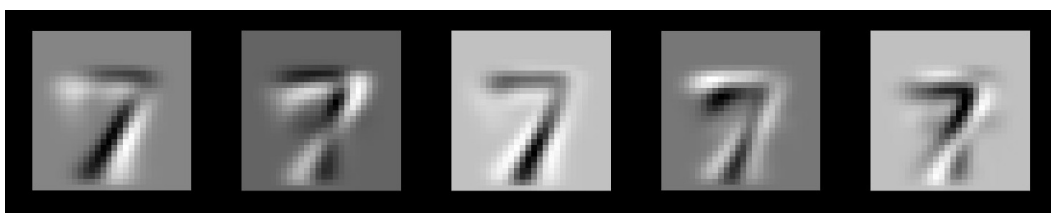
Digit 0



Digit 3



Digit 6



Digit 7